Jiangtao Xie

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Educational background

➢**Ocean University of China (A-Class of "Double First-Class" Institution, 985 Institution)**

Department of Information Science and Engineering **Electronic Information Engineering Major**

➢**Overall academic performance and ranking: 84.68/100 (rank:24/111)** **IELTS：6.5 Reading:7.0 Writing:7.0**

➢**Course grade： High-resolution spectral analysis（97）、Deep learning（95.6）、Advanced language programme design(93)**
Electronic circuit design（95）、Fundamentals of Circuit Analysis（91）、Artificial intelligence（91.5）、Machine vision(100)
Electronic Skills Training（95）、Quantum mechanics and quantum information theory（90）、Acoustic measurement (93)
Method of mathematical physics（91.6）、Computer aided drafting（90）、Learn Digital Economics with AI (100)
International Order and Treaties (100)、Metalworking Practice(95.7)、Quantum Computing Competition（A）

Scholarship: Scholarship of Ocean University of China*3、BYD Enterprise Scholarship

High School Academic Competitions：CHSBO 2022 Provincial First Prize | CPho 2022 Provincial Second Prize

Scientific research experience

Design of a ground-to-air collaborative inspection robot for chemical plants with active suspension system
Provincial-level Innovation Project

Mechanical Structure Design: Designed an electric-cylinder-driven quadruped robot chassis and completed 3D modeling and motion simulation in SolidWorks.
Active Suspension Control: Developed a cascade PID control system based on MPU6050 attitude feedback, enabling the robot to maintain stability on 15° slopes and overcome 20 cm obstacles, with body inclination kept below 3°.Electronic Control System: Designed the main control board based on the STM32F103VET6, incorporating power management and CAN, SPI, and I²C communication interfaces.
Multi-Sensor Integration: Integrated the MQ-2 gas sensor, SHT30 temperature and humidity sensor, and RPLIDAR A1 LiDAR for environmental monitoring and perception.
High-Speed Data Acquisition: Designed a custom PCB and signal-conditioning circuits to achieve multi-source sensor data acquisition at a sampling rate of at least 100 Hz.
Autonomous Navigation and Obstacle Avoidance: Implemented SLAM-based localization, A* path planning, and Dynamic Window Approach navigation on the ROS platform, achieving a positioning error within ±10 cm.
Ground – Air Collaborative Positioning: Optimized the fusion of visual sensing and UWB positioning to enable precise docking between the UAV and the ground robot, with a docking error of less than 3 cm.

Numerical simulation of sedimentary sequence evolution and hydrocarbon migration and accumulation model under the control of strike-slip pull-apart basin formation and inversion structure

University-level Innovation and Entrepreneurship Program

Multiphysics Algorithm Development: Independently developed a three-dimensional two-way sedimentary－tectonic coupling framework by integrating ASPECT and Badlands, demonstrating strong capabilities in numerical algorithm design, multiphysics coupling, and scientific software development.
Finite-Element Modeling: Built a visco-elastic-plastic finite-element model incorporating dynamic boundary conditions, realistic crustal thermal structures, fault geometry, and material heterogeneity.
High-Performance Numerical Simulation: Implemented full-lifecycle simulations of the dynamic feedback between deep tectonic deformation and shallow sedimentary loading, with experience in large-scale computational modeling, parameter configuration, convergence analysis, and simulation stability control.Geodynamic
Data Analysis: Analyzed stress migration, deformation patterns, sedimentary evolution, and fault-controlled basin development, demonstrating strong skills in extracting geological mechanisms from complex simulation outputs.
Scientific Visualization: Processed and visualized three-dimensional simulation results, including stress fields, velocity fields, deformation patterns, sediment thickness, and basin morphology.
Machine Learning Integration: Applied machine learning together with coupled numerical simulation to develop a paleotopographic inversion method for marginal-sea hydrocarbon basins.One patent application has been approved.
Geoscientific AI and Inversion: Contributed to two granted invention patents on multi-source cross-modal geostress prediction and machine-learning -based paleotopographic inversion, with experience in geological data fusion, intelligent modeling, inverse-problem solving, and computational geoscience.

Underwater cleaning robot for ship propellers

University-level Innovation and Entrepreneurship Program

Electronic System Integration: Led the design of a Jetson Nano-based distributed control system and independently developed a multi-channel power management PCB, ensuring stable power supply, low-noise operation, and reliable integration of complex underwater robotic hardware.
Deep Learning-Based Vision Development: Developed a real-time underwater image enhancement model to address turbidity, color distortion , uneven illumination, and cavitation-jet interference.
Edge AI Deployment: Optimized and lightweighted the model for real-time high-definition video processing on Jetson Nano, demonstrating strong capabilities in embedded AI, model optimization, and hardware – software co-design.
System Performance Improvement: Significantly enhanced underwater visual perception and provided reliable technical support for remote monitoring and precise robot control.

Fabrication and Application of Visible-Transparent Infrared Organic Photodetectors (OPDs)
Research Internship at the University of Science and Technology of China
Device Research and Optimization: Participated in Prof. Jun Liu's resonant boron-nitrogen bond-based SWIR detector research; studied 1000-3000 nm photoelectric response and optimized fabrication to retain transparency and improve infrared sensitivity.
Image Processing and Device Integration: Processed infrared imaging data, evaluated image quality, and contributed to integrating flexible, visible-transparent organic photodetectors with image acquisition and processing.

Metagenomic analysis reveals Northwest Pacific Ocean as a reservoir and evolutionary hub of antibiotic resistance genes
Environmental Pollution | SCI JCR Q1 | Third Author
Bioinformatics Pipeline: Integrated Trimmomatic, MEGAHIT, Prodigal and DIAMOND/CARD for quality control, assembly, gene prediction and antibiotic resistance gene annotation.
Statistical Modeling: Built Spearman co-occurrence networks with bootstrap validation; applied zero replacement and centered log-ratio transformation to address compositional bias.
Functional Analysis: Used KEGG/eggNOG annotation to characterize efflux-pump-mediated resistance and interpret marine resistance gene patterns.

Subject competition experience

2026 "Challenge Cup" China Youth Science and Technology Innovation Challenge-Based Competition — National First Prize, "Research and Applications of Bidirectional Synergy between Quantum Technologies and Artificial Intelligence"
Tensor-Product Quantum Convolutional Neural Networks for Image Classification
Architecture and Modeling: Contributed to a two-layer, multichannel QCNN using amplitude encoding and tensor-product parameter sharing, with complementary Z/X readout and classical pooling/re-encoding.
Simulation and Evaluation: The project achieved 98.36%, 87.53% and 85.84% accuracy on MNIST, FashionMNIST and PneumoniaMNIST, respectively, in state-vector simulations across three random seeds.
Compression and Robustness: Reduced the MNIST model from 3,640 to 520 parameters (85.71%) while retaining 96.06% accuracy; evaluated trainability, readout ablations and gate-level noise robustness.

National Undergraduate Competitionon Embedded Chips and Systems Design (National ThirdPrize)
AI Algorithm Design: Developed a hybrid recognition architecture combining a lightweight LeNet neural network with font-matching algorithms to accurately recognize Chinese characters, letters, and numbers.
Hardware Implementation: Used Verilog HDL to complete the full hardware design process, including image preprocessing, license plate localization, character segmentation, and recognition.
Hardware – Software Co-Design: Efficiently mapped the recognition algorithms onto FPGA through modular architecture and pipelining, demonstrating strong capabilities in AI algorithm optimization, digital circuit design, and embedded system integration.
Performance Optimization: Improved real-time processing capability and energy efficiency, enabling reliable deployment of the complete license plate recognition pipeline on resource-constrained FPGA hardware.

International Genetically Engineered Machine Competition (Global Silver Award, Hardware Compliance Gold Award)
Robotic System Development: Built a ROS- and Gazebo-based digital twin for wound contour recognition, manipulator motion, and 3D path planning, and developed a custom inverse-kinematics solver for closed-loop simulation and physical-system deployment.
Two-Stage Visual Perception: Developed a coarse-to-fine wound recognition pipeline in which YOLOv8-seg first rapidly localizes the wound region, followed by FCN-based fine-grained segmentation within the detected area to improve boundary accuracy, detail preservation, and robustness across different skin tones.
AI Agent and Computational Modeling: Integrated AI agents into the dry-lab workflow for literature retrieval, antimicrobial peptide generation, property prediction, explainable ranking, result comparison, and experimental data interpretation, connecting computational screening with wet-lab validation.Conducted **molecular dynamics simulations** to quantitatively analyze hydrogel gelation mechanisms and peptide– carrier interactions
Multimodal System Integration: Combined polar-coordinate path planning, motion-control optimization, low-cost depth estimation, temperature and pH sensing, and IoT transmission, demonstrating strong capabilities in robotics, computer vision, AI modeling, automation, and hardware – software integration.

International Directed Evolution Contest (Best Molecular EvolutionResults | BestSoftware Design | Best Machine Learning Driven Directed Program Nomination)
Analysis: Analyzed enzyme structures and key residues to guide mutation-site selection and candidate design.
Modeling: Built machine learning models to predict and rank promising enzyme variants using multidimensional amino-acid features.
Validation: Integrated computational screening with protein expression, purification, and activity evaluation to validate optimized mutants.

CCF Quantum Computing Programming Challenge - Original Source Cup (National Second Prize)
Quantum Theoretical Modeling: Derived Hamiltonian matrices, parameterized quantum states, energy expectation functions, and optimal variational parameters, transforming quantum-mechanical formulations into executable optimization models.
Quantum Circuit Implementation: Implemented quantum algorithms in PyQPanda3, including entanglement preparation, quantum teleportation, measurement-based control, conditional correction, probability analysis, and fidelity verification.
Variational Quantum Architecture: Designed an 8-qubit baseline VQC and a lightweight 2-qubit Data Re-uploading classifier, enabling complete encoding of 16-dimensional medical features with only 6 trainable quantum parameters.
Distillation and Compression: Integrated knowledge distillation, loss regularization, circuit compression, and hyperparameter optimization, reducing qubit usage by 75% and trainable quantum parameters by 92.5% while maintaining effective classification performance.

National College Students' Mathematical Modeling Contest (Provincial Second Prize)
National College Students' Intelligent Vehicle Competition (Provincial Second Prize)
China College Students ' Engineering Practice and Innovation Ability Competition (Provincial Third Prize)
Mathematical Contest In Modeling 2026 Honorable Mention

Other information

- Engineering ability : skilled use of Solidwork, Jialichuang EDA, MATLAB, Fushion360, Dify, Coze, Linux system, supercomputing, GPU cluster use, training experience, large model deployment fine-tuning experience, use ROS system to control the mainstream manipulator, mainstream single-chipmicrocomputer use experience.